Ramalingam Naidu Allu

Third Year Undergraduate
Department of Computer Science and Engineering
Indian Institute of Technology, Kanpur

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2024-Present	B.Tech	Indian Institute of Technology Kanpur	**6.3/10**
2024	CBSE(XII)	Sri Chaitanya Vidya Niketan , Madhurawada	95.6%
2022	CBSE(X)	Jawahar Navodaya Vidyalaya , Srikakulam	93.4%

Key Projects

Qwik | ***Course Project, CS253*** | ***Prof. Indranil Saha*** *Jan'26 - Apr'26*

- Collaborated within a **10-member development team** to design, develop, and deploy a full-stack digital canteen platform utilizing **React**, **Node.js**, and **Express.js** to streamline role-based access for campus customers, merchants, and super administrators.
- Implemented secure user authentication using **JWT** sessions, **email-based OTP verification**, and **bcrypt** password hashing.
- Developed **optimized RESTful API endpoints** and schema architectures within **MongoDB** to efficiently handle dynamic menu modifications, automated user profiles, real-time live order queues, and operational analytics for canteen management.
- Integrated **Stripe payment** gateways alongside robust server-side transaction validation to secure the order lifecycle.

GanForge | ***Association of Computing Activities*** *May'25 - June'25*

- Developed **GanForge, a generative AI framework** leveraging CycleGAN architectures to perform unsupervised image-to-image translation across unpaired datasets, mapping complex visual features to adapt real-world photographs into Monet-style artwork.
- Implemented adversarial, cycle-consistency, and identity loss functions in PyTorch to stabilize training and maintain image content.
- Utilized **OpenCV** to engineer automated computer vision pipelines, applying custom convolutional kernels, complex matrix filters, edge detection, and thresholding techniques to reliably extract structural features and image contours prior to model training.
- Evaluated translation performance across multi-domain datasets using structural metrics to verify visual output quality.

Memory-Efficient Versioned File Indexer | ***C++ Project, CS253*** | ***Prof. Indranil Saha*** *Feb'26 - Mar'26*

- Developed a memory-efficient word-level indexing system for large text files utilizing fixed-size buffered file processing architectures.
- Implemented **versioned indexing**, **word-frequency lookup**, **top-K retrieval**, and analytical cross-version comparison tools while simultaneously ensuring that total memory consumption metrics remain entirely independent of the input file size.
- Applied templates, inheritance, runtime polymorphism, exception handling, and function overloading inside a modular framework.
- Optimized large-scale text processing through chunk-based reading and high-efficiency hash-map indexing frameworks.

Mini-MIPS Simulator | ***Course Project, CS220*** | ***Prof. Mainak Chaudari*** *Mar'26 - Apr'26*

- Designed and implemented a single-cycle MIPS processor in Verilog, replicating the **5-stage flow**: IF, ID, EX, MEM, and WB.
- Developed and tested support for core MIPS instructions, including arithmetic, logical, memory, and branching operations.
- Integrated **IEEE 754 single-precision** floating-point multiplication into the ALU alongside existing integer arithmetic logic.
- Validated processor functionality by executing insertion sort and verifying correct instruction flow using Vivado tools.

Technical Skills

- **Skills :** Data Structures and Algorithms (DSA) | Web Development | Object-Oriented Programming (OOP)
- **Programming Languages :** C | C++ | Python | HTML | CSS | JavaScript | MIPS | SQL | Verilog HDL | Bash | LaTeX
- **Database & Tools :** MongoDB | PostgreSQL | Git | GitHub | Jupyter Notebook | VS Code
- **Web Frameworks & Libraries :** React.js | Node.js | Express.js | RESTful APIs | NumPy | Pandas | Matplotlib | PyTorch
- **Competitive Programming :** Maximum rating of **1444**(Handle : **ColdAscent**) on **codeforces(Specialist)**

Relevant Courses

Fundamentals of Computing	Data Structures & Algorithms	Logic & Probability in Computer Science
Discrete Mathematics	Introduction to Machine Learning	Introduction to Electronics
Computer Organization	Software Development and Operations	Linear Algebra & Differential Equations

Extra-Curricular Activities

- **Won Gold Medal** in **Hockey** at **Freshers' Inferno 2024**, held at **IIT Kanpur**, demonstrating teamwork and discipline
- **Completed National Cadet Corps (NCC) training** during the first year of undergraduate studies, successfully enduring rigorous physical conditioning drills while systematically developing essential traits in leadership, discipline, and group teamwork.